Two-way ANOVA with balanced design w/o replicate — 29.10.2024 LSM

Implementation of Tukey's test for additivity:

Standard ANOVA with replicates → implementation in R by using aov function

Use as.factor to convert levels to categorical predictor

Can use package "additivity test"

Enter the data in the form of a matrix

R Code: tukey.test (insurance, alpha = 0.05) ...

For computation by hand

$$\left.\begin{array}{l} \text{in addition to } \bar{Y}_{i\cdot}, \bar{Y}_{\cdot j}, \bar{Y}_{\cdot\cdot} \\ SSA, SSB, SSAB = SSTo - SSA - SSB \end{array}\right\}$$

$$SSAB = \hat{\theta}^2 \sum \hat{\alpha}_i^2 \sum \hat{\beta}_j^2$$

$$\hat{\theta} = \frac{\sum_i \sum_j (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j)\hat{\alpha}_i \hat{\beta}_j}{\sum_i \hat{\alpha}_i^2 \sum_j \hat{\beta}_j^2} = \frac{\sum_i \sum_j \hat{\alpha}_i \hat{\beta}_j Y_{ij}}{\sum_i \hat{\alpha}_i^2 \sum_j \hat{\beta}_j^2}$$

$$\left\{\begin{array}{l} SSA = b \sum_i \hat{\alpha}_i^2 \\ SSB = a \sum_j \hat{\beta}_j^2 \end{array}\right.$$

check that,

$$\sum_i \sum_j \hat{\alpha}_i \hat{\beta}_j Y_{ij}$$

$$\left.\begin{array}{l} = \sum_i \sum_j \bar{Y}_{i\cdot} \bar{Y}_{j\cdot} Y_{ij} \\ - (SSA + SSB)\bar{Y}_{\cdots} - ab\,\bar{Y}_{\cdots}^3 \end{array}\right\}$$

Nested ANOVA model:

Structure of the data is necessarily nested.

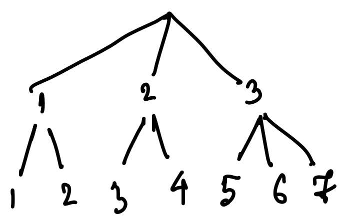
① A study on production process involved collecting data on several plants within each plant, a certain number of operators run the machines.

A sample data may look like the following

plant \ operator	1	2	3	4	5	6	7
1	x	x					
2			x	x			
3					x	x	x

Statistical Model:

$$Y_{ijk} = \mu + \alpha_i + \nu_{ij} + \varepsilon_{ijk}$$



$i = 1, \ldots, a$
$j = 1, \ldots, b_i$
$k = 1, \ldots, m_{ij}$

ν_{ij}: specific effect of j^{th} level of factor B within i^{th} level of factor A

② Study on performance of soldiers in an army unit

A divisional commander may conduct a study about performance of different units within the division in accomplishing certain tasks

Identifiability constraints

(a) "Standard" constraint: $\sum_{i=1}^{a} \alpha_i = 0$

$$\sum_{j=1}^{b_i} \gamma_{ij} = 0 \text{ for all } i$$

(b) "Orthogonal-inducing" constraint

$$\sum_{j=1}^{b_i} m_{ij} \gamma_{ij} = 0 \quad \text{for } j = 1, \dots, a$$

$$\sum_{i=1}^{a} n_i \alpha_i = 0 \quad \text{where } n_i = \sum_{j=1}^{b_i} m_{ij}$$

Motivation for the constraints (b)

Find BLUE for μ, $\{\alpha_i\}$, $\{\gamma_{ij}\}$

$$\mu = \frac{1}{n} \sum_i \sum_j m_{ij} \mu_{ij} \quad \text{where } n = \sum_i \sum_j m_{ij} = \sum_i n_i$$

$$\alpha_i = \frac{1}{n} \sum_{j=1}^{b_i} m_{ij} \mu_{ij} - \mu$$

$$\gamma_{ij} = \mu_{ij} - \mu - \alpha_i$$

LSE for μ_{ij}

$$\hat{\underline{\mu}} = \underset{\mu_{ij}}{\operatorname{argmin}} \sum_i \sum_j \sum_k (Y_{ijk} - \mu_{ij})^2$$

$$\hat{\mu}_{ij} = \overline{Y_{ij\cdot}} \qquad \hat{\alpha}_i = \overline{Y_{i\cdot\cdot}} - \overline{Y_{\cdot\cdot\cdot}} \qquad \hat{\gamma}_{ij} = \overline{Y_{ij\cdot}} - \overline{Y_{i\cdot\cdot}}$$

Under constraints ⓑ BLUE for $\mu, \{\alpha_i\}, \{\gamma_{ij}\}$ are also LSE for those parameters

Brief calculation: Show that $\begin{cases} \sum n_i \hat{\alpha}_i = 0 \\ \sum m_{ij} \hat{\gamma}_{ij} = 0 \text{ for all } i \end{cases}$

and under constraint ⓑ

$$\sum\sum\sum (Y_{ijk} - \mu - \alpha_i - \gamma_{ij})^2$$

$$= \sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y_{ij\cdot}})^2 + n(\mu - \overline{Y_{\cdots}})^2$$

$$+ \sum_i n_i (\alpha_i - (\overline{Y_{i\cdot\cdot}} - \overline{Y_{\cdots}}))^2 + \sum_i \sum_j m_{ij} (\gamma_{ij} - (\overline{Y_{ij\cdot}} - \overline{Y_{i\cdot\cdot}}))^2$$

Set $\mu = \overline{Y_{\cdots}}$, $\alpha_i = 0$, $\gamma_{ij} = 0$ to obtain the ANOVA decomposition

$$\underbrace{\sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y_{\cdots}})^2}_{SSTo} = \underbrace{\sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y_{ij\cdot}})^2}_{SSE} + SSA + \underbrace{\sum_i \sum_j m_{ij} \hat{\gamma}_{ij}^2}_{SSB(A)}$$

$$SSA = \sum n_i \hat{\alpha}_i^2 = \sum n_i (\overline{Y_{i\cdot\cdot}} - \overline{Y_{\cdots}})^2$$

$$SSB(A) = \sum_i \sum_j m_{ij} (\overline{Y_{ij\cdot}} - \overline{Y_{i\cdot\cdot}})^2$$

SS due to specific effect of factor B nested within factor A

Notice that

$\hat{\mu}, \{\hat{\alpha}_i\}, \{\hat{\gamma}_{ij}\}$ are uncorrelated

$Cov(\hat{\mu}, \hat{\alpha}_i) = 0$

$Cov(\hat{\mu}, \hat{\gamma}_{ij}) = 0$

$Cov(\hat{\alpha}_i, \hat{\gamma}_{ij}) = 0$

df

$$df(SSTo) = df(SSE) + df(SSA) + df(SSB(A))$$

$$n-1 \qquad n - \sum_{i=1}^{a} b_i = \sum_{i=1}^{a} (n_i - b_i) = n - a\bar{b} \qquad a-1 \qquad \sum_{i=1}^{a} (b_i - 1) = a\bar{b} - a$$

Def. $\bar{b} = \frac{1}{a} \sum_{i=1}^{a} b_i$

Test for main effects of factor A

$H_0^A: \alpha_1 = \dots = \alpha_a = 0$ vs. H_1^A: not all α_i's are zero

$$F^A = \frac{MSA}{MSE} = \frac{SSA/a-1}{SSE/n-a\bar{b}} \overset{H_0}{\sim} F_{a-1,\, n-a\bar{b}}$$

Test for specific effects of factor B

$H_0^B: \nu_{ij} = 0 \;\; \forall j \; \forall i$ $\quad H_1^B$: not all ν_{ij}'s are zero

$$F^B = \frac{MSB(A)}{MSE} = \frac{SSB(A)/a(\bar{b}-1)}{SSE/n-a\bar{b}} \overset{H_0}{\sim} F_{a(\bar{b}-1),\, n-a\bar{b}}$$

(a) "Standard constraints"

$\sum_i \alpha_i = 0, \; \sum_j \nu_{ij} = 0$ for all i

$$\mu_{ij} = \mu + \alpha_i + \nu_{ij} \qquad \sum_i (\mu + \alpha_i) = a\mu$$

$$\sum_j \mu_{ij} = b_i\mu + b_i\alpha_i + 0 \qquad \mu = \frac{1}{a}\sum_i (\mu + \alpha_i)$$

$$\mu + \alpha_i = \frac{1}{b_i}\sum_{j=1}^{b_i} \mu_{ij} = \bar{\mu}_{i\cdot} \qquad = \frac{1}{a}\sum_i \frac{1}{b_{ij}}\sum_{j=1}^{b_i}\mu_{ij}$$

$$\alpha_i = \frac{1}{b}\sum_{j=1}^{b_i} \mu_{ij} - \frac{1}{a}\sum_{i'} \frac{1}{b_{i'}} \sum_{j=1}^{b_{i'}} \mu_{i'j}$$

$$\nu_{ij} = \mu_{ij} - (\mu + \alpha_i)$$

Formulation of the estimation problem as a regression problem

$$\underset{n\times 1}{\underline{Y}} = \mu \underset{n\times 1}{\underline{1}_n} + \underset{n\times a}{X_A} \underset{a\times 1}{\underline{\alpha}} + \sum_{i=1}^{a} \underset{n\times b_i}{X_B^{(i)}} \underset{b_i\times 1}{\underline{\gamma}_i} + \underline{\varepsilon}$$

when $\underline{\alpha} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_a \end{pmatrix}$, $\underline{\gamma}_i = \begin{pmatrix} \gamma_{i1} \\ \vdots \\ \gamma_{ib_i} \end{pmatrix}$

Use constraint (a)

$$\alpha_a = -\sum_{i=1}^{a-1} \alpha_i \quad \text{and} \quad \gamma_{ib_i} = -\sum_{j=1}^{b_i-1} \gamma_{ij}$$

Transform the design matrix:

Define $\alpha^- = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_{a-1} \end{pmatrix}$ $\quad \gamma_i^- = \begin{pmatrix} \gamma_{i1} \\ \vdots \\ \gamma_{i(b_i-1)} \end{pmatrix}$

$$\tilde{X}_{A,i} = X_{A,i} - X_{A,a}$$

$$\tilde{X}^{(i)}_{B,j} = X^{(i)}_{B,j} - X^{(i)}_{B,b_i}$$

$$\underline{Y} = \mu \underline{1} + \tilde{X}_A \underline{\alpha}^- + \sum_{i=1}^{a} \tilde{X}_B^{(i)} \underline{\gamma}_i^- + \underline{\varepsilon}$$

$$\left[\underset{1}{1} : \underset{a-1}{\tilde{X}_A} : \underset{b_1-1}{\tilde{X}_B^{(1)}} : \dots : \underset{b_a-1}{\tilde{X}_B^{(a)}} \right]$$